

GENERAL EQUILIBRIUM THEORY

Research Master in Economics at Tilburg University

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Chapter 1

Introduction

This set of notes accompanies the general equilibrium theory course in the Research Master in Economics at Tilburg University.

Chapter 2

The Private Ownership Economy

In this chapter we will give the definitions of an economy and a private ownership economy for this course. We will use a recurring example of a one-person and one-firm economy illustrate this. We will define Pareto optimality, a minimal condition for a plan of production and consumption to be efficient (though not necessarily fair), and the notion of a Walrasian equilibrium.

2.1 Definition of an Economy

There are $I > 0$ consumers, $J > 0$ firms and L commodities in the economy. Each consumer i is characterized by a consumption set $X_i \subset \mathbb{R}^L$ and a complete and transitive preference relation $\succeq_i$ defined over X_i . Each firm j is characterized by a nonempty and closed production set $Y_j \subset \mathbb{R}^L$. There is an initial endowment $\bar{\omega} = (\bar{\omega}_1, \dots, \bar{\omega}_L) \in \mathbb{R}_+^L$ of each good. This is how much of each good ℓ exists in the economy before any production takes place. An *economy* can be summarized by:

$$\mathcal{E} = \left(\{ (X_i, \succeq_i) \}_{i=1}^I, \{ Y_j \}_{j=1}^J, \bar{\omega} \right)$$

Robinson Crusoe Example:

In this chapter we will frequently return to a commonly-used example in Economics. Robinson Crusoe is the eponymous character from a book by Daniel Defoe from 1719. He is a lone castaway on the Island of Despair. On this island there are two goods: coconuts and hours of leisure ($L = 2$). Crusoe is the only consumer on the island ($I = 1$) and a coconut-producing firm is the only firm ($J = 1$). Crusoe can consume $x_{11} \geq 0$ coconuts but at most 24 hours of leisure per day. Therefore his consumption set is $X_1 = \mathbb{R}_+ \times [0, 24]$. Crusoe has preferences over X_1 given by $\succeq_1$. The firm's

production technology is producing coconuts from leisure:

$$\mathcal{Y}_1 = \{(y_{11}, y_{21}) \in \mathbb{R}^2 : y_{11} \leq f(-y_{21})\}$$

The firm technology is essentially Crusoe using his leisure to climb trees and pick coconuts. The initial endowment is no (ready-to-eat) coconuts and 24 hours of leisure, so $\bar{\omega} = (0, 24)$.

2.2 Feasible Allocations

An *allocation* $(\mathbf{x}, \mathbf{y}) = (\mathbf{x}_1, \dots, \mathbf{x}_I, \mathbf{y}_1, \dots, \mathbf{y}_J)$ is a specification of a consumption vector $\mathbf{x}_i \in \mathcal{X}_i$ for each consumer $i = 1, \dots, I$ and production vector $\mathbf{y}_j \in \mathcal{Y}_j$ for each firm $j = 1, \dots, J$.

The allocation $(\mathbf{x}_1, \dots, \mathbf{x}_I, \mathbf{y}_1, \dots, \mathbf{y}_J)$ is *feasible* if

$$\sum_{i=1}^I x_{\ell i} = \bar{\omega}_{\ell} + \sum_{j=1}^J y_{\ell j} \quad \forall \ell = 1, \dots, L$$

We write the feasibility condition with equality because we typically assume firms have free disposal. Under free disposal, if there is an undesirable good and some quantity of the endowment is leftover, then the firm can destroy it to obtain feasibility.

Robinson Crusoe Example:

On the Island of Despair, an allocation $(\mathbf{x}_1, \mathbf{y}_1) \in \mathcal{X}_1 \times \mathcal{Y}_1$ is feasible if:

$$\begin{aligned} \text{Coconuts: } x_{11} &= 0 + y_{11} && \text{with } x_{11} \geq 0 \\ \text{Leisure: } x_{21} &= 24 + y_{21} && \text{with } x_{21} \in [0, 24] \end{aligned}$$

The total quantity of coconuts consumed must equal the total quantity produced by the firm (or picked from the tree). The total quantity of leisure consumed equals 24 minus the amount of labor used by the firm. Because y_{21} is an input, its value will be negative.

We can combine feasibility with the production technology to write these only in terms of the consumption goods:

$$x_{11} = y_{11} \leq f(-y_{21}) = f(24 - x_{21})$$

2.3 Pareto Optimality

Denote the set of feasible allocations by:

$$\mathcal{A} = \left\{ (\mathbf{x}, \mathbf{y}) \in \prod_{i=1}^I \mathcal{X}_i \times \prod_{j=1}^J \mathcal{Y}_j : \sum_{i=1}^I \mathbf{x}_i = \bar{\omega} + \sum_{j=1}^J \mathbf{y}_j \right\} \subset \mathbb{R}^{L(I+J)}$$

A feasible allocation $(\mathbf{x}, \mathbf{y})$ is *Pareto optimal* if there is no other feasible allocation $(\mathbf{x}', \mathbf{y}') \in \mathcal{A}$ that *Pareto dominates* it, that is, if there is no feasible allocation $(\mathbf{x}', \mathbf{y}') \in \mathcal{A}$ such that $\mathbf{x}'_i \succeq_i \mathbf{x}_i$ for all i and $\mathbf{x}'_i \succ_i \mathbf{x}_i$ for some i .

An allocation is Pareto optimal if there is no way to make someone better off without making someone else worse off. Pareto optimality does not have any notion of fairness or equity. It is merely a minimum condition for an allocation to be efficient. Without it, we are leaving money on the table.

Robinson Crusoe Example:

Assume Crusoe's preferences over $\mathcal{X}_1$ are continuous, convex and monotone and can be represented by:

$$u_1(x_{11}, x_{21}) = x_{11}x_{21}$$

Assume the firm's production technology is:

$$\mathcal{Y}_1 = \{(y_{11}, y_{21}) : 0 \leq y_{11} \leq \sqrt{-y_{21}} \text{ and } y_{21} \in [-24, 0]\}$$

This means that z hours of labor in a day can product $\sqrt{z}$ coconuts.

Let's find the set of Pareto optimal allocations on the island. Because $I = 1$, a Pareto optimal allocation solves the following constrained maximization problem:

$$\max_{(x_{11}, x_{21})} x_{11}x_{21}$$

subject to:

- $x_{11} \geq 0$ and $x_{21} \in [0, 24]$ (consumption set)
- $y_{11} \leq \sqrt{-y_{21}}$ (production technology)
- $x_{11} = y_{11}$ and $x_{21} = 24 + y_{21}$ (feasibility)

Combining the technology and feasibility constraints:

$$x_{11} = y_{11} \leq \sqrt{-y_{21}} = \sqrt{24 - x_{21}}$$

Because utility is increasing in both goods, there is no disposal and so $x_{11} = \sqrt{24 - x_{21}}$. Combining this with the constraint $x_{21} \in [0, 24]$ allows us to rewrite the maximization problem as:

$$\max_{x_{21} \in [0, 24]} \left(\sqrt{24 - x_{21}} \right) x_{21}$$

This problem has a unique solution at $x_{21} = 16$, which implies the unique Pareto allocation in this economy is:

$$\begin{aligned} (x_{11}, x_{21}) &= (\sqrt{8}, 16) \\ (y_{11}, y_{21}) &= (\sqrt{8}, -8) \end{aligned}$$

Crusoe works for 8 hours and enjoys 16 hours of leisure. His labor produces $\sqrt{8}$ coconuts which he consumes.

2.4 Private Ownership Economies

In a standard private ownership economy, a market exists for each good and consumers and firms are price takers. Prices are given by $\mathbf{p} \in \mathbb{R}^L$. Each consumer i owns an initial endowment $\omega_i \in \mathbb{R}_+^L$, where $\bar{\omega}_\ell = \sum_{i=1}^I \omega_{\ell i} \forall \ell$. Each consumer i has a claim to a share $\theta_{ij} \in [0, 1]$ of the profits of firm j , where $\sum_{i=1}^I \theta_{ij} = 1 \forall j$. Consumer i 's budget set with price vector $\mathbf{p} \in \mathbb{R}^L$ is then:

$$\mathcal{B}_i(\mathbf{p}) = \left\{ \mathbf{x}_i \in \mathcal{X}_i : \mathbf{p} \cdot \mathbf{x}_i \leq \mathbf{p} \cdot \omega_i + \sum_{j=1}^J \theta_{ij} \mathbf{p} \cdot \mathbf{y}_j \right\}$$

A *private ownership economy* can be summarized by:

$$\mathcal{E}_p = \left(\{(\mathcal{X}_i, \succeq_i)\}_{i=1}^I, \{\mathcal{Y}_j\}_{j=1}^J, \{(\omega_i, \theta_{i1}, \dots, \theta_{iJ})\}_{i=1}^I \right)$$

Robinson Crusoe Example:

There is a market for coconuts and labor. The price of a coconut is p_1 and the price of one hour of leisure is p_2 . Crusoe owns the full initial endowment: $\omega_1 = \bar{\omega} = (0, 24)'$. Crusoe owns a 100% share in the coconut firm: $\theta_{11} = 1$. The firm hires Crusoe for labor and produces coconuts. The firm then sells Crusoe the coconuts. Crusoe's budget set is then all $\mathbf{x}_i \in \mathcal{X}_i$ that satisfy:

$$\underbrace{p_1 x_{11} + p_2 x_{21}}_{\text{Cost of consumption}} \leq \underbrace{24 p_2}_{\text{Value of endowment}} + \underbrace{1 \times [p_1 y_{11} + p_2 y_{21}]}_{\text{Share of firm's profits}}$$

2.5 Walrasian Equilibrium

Given a private ownership economy $\mathcal{E}_p$, an allocation $(\mathbf{x}^*, \mathbf{y}^*)$ and a price vector $\mathbf{p} \in \mathbb{R}^L$ constitute a *Walrasian equilibrium* if:

- (i) For every j , $\mathbf{y}_j^*$ maximizes profits in $\mathcal{Y}_j$; that is $\mathbf{p} \cdot \mathbf{y}_j \leq \mathbf{p} \cdot \mathbf{y}_j^* \forall \mathbf{y}_j \in \mathcal{Y}_j$.
- (ii) For every i , $\mathbf{x}_i^*$ is maximal for $\succeq_i$ in the budget set $\mathcal{B}_i(\mathbf{p})$
- (iii) $\sum_{i=1}^I \mathbf{x}_i^* = \bar{\omega} + \sum_{j=1}^J \mathbf{y}_j^*$

Robinson Crusoe Example:

We will now solve for the Walrasian equilibrium on the Island of Despair. First we argue that we must have $\mathbf{p} \gg \mathbf{0}$ in equilibrium. There cannot be an equilibrium with $p_2 = 0$ because then the firm would want to use an infinite amount of input 2 (labor), which the consumer would not be willing to supply (Crusoe will not want to work for free). There also cannot be an equilibrium with $p_1 = 0$ because then the firm would not produce any quantity of good 1 (coconuts) whereas Crusoe would demand an infinite quantity.

We now begin with the firm's problem. The firm solves:

$$\max_{\{\mathbf{y}_1 \in \mathcal{Y}_1\}} p_1 y_{11} + p_2 y_{21}$$

Provided $\mathbf{p} \gg \mathbf{0}$, the firm will optimally set $y_{11} = \sqrt{-y_{21}}$. With this, the problem becomes:

$$\max_{y_{21} \leq 0} p_1 \sqrt{-y_{21}} + p_2 y_{21}$$

In an interior solution, the first-order condition is $-\frac{p_1}{2\sqrt{-y_{21}}} + p_2 = 0$. This implies $y_{21} = -\frac{p_1^2}{4p_2^2}$ and $y_{11} = \frac{p_1}{2p_2}$. The firm's profit function is then:

$$\pi_1(\mathbf{p}) = \frac{p_1^2}{2p_2} - \frac{p_1^2}{4p_2} = \frac{p_1^2}{4p_2}$$

We show the firm's optimal production plan graphically in Figure 2.1. The optimal choice of y_{21} is when the firm's isoprofit line is tangent to the production set $\mathcal{Y}_1$. We can find the firm's isoprofit line by solving $\pi_1 = p_1 y_{11} + p_2 y_{21}$ for y_{11} . This is given here by $\frac{\pi_1}{p_1} - \frac{p_2}{p_1} y_{21}$. The consumer solves the following problem:

$$\max_{\{x_{11} \geq 0, x_{21} \in [0, 24]\}} x_{11} x_{21}$$

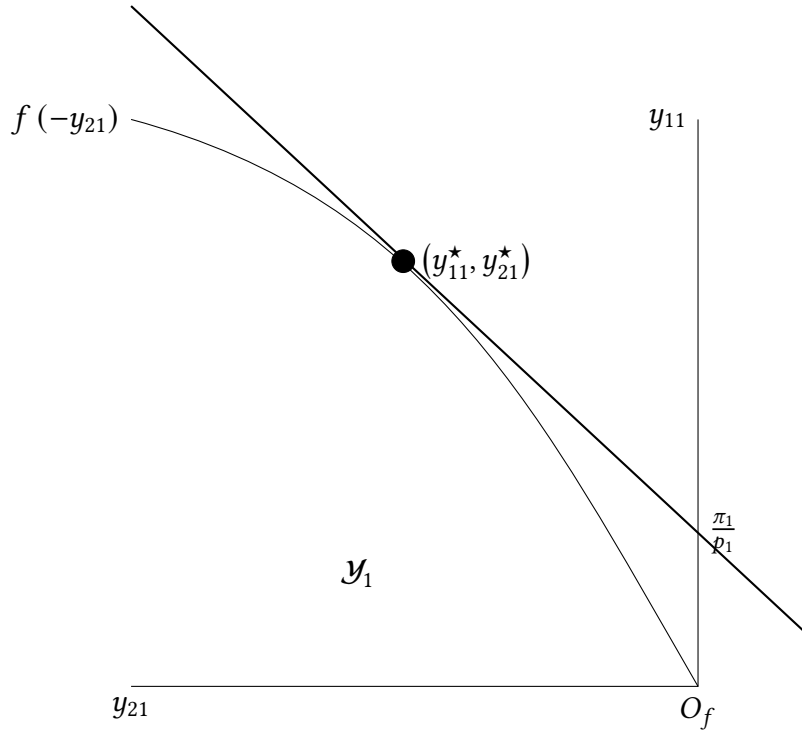


FIGURE 2.1: The firm's optimal production plan in the Robinson Crusoe economy.

subject to the budget constraint:

$$p_1 x_{11} + p_2 x_{21} \leq 24p_2 + \frac{p_1^2}{4p_2}$$

Because utility is increasing in both goods, the budget constraint will bind at the optimal solution:

$$x_{11} = \frac{p_2}{p_1} (24 - x_{21}) + \frac{p_1}{4p_2}$$

The problem is then:

$$\max_{\{x_{21} \in [0, 24]\}} \left(\frac{p_2}{p_1} (24 - x_{21}) + \frac{p_1}{4p_2} \right) x_{21}$$

In an interior solution, the first-order conditions is:

$$\frac{p_2}{p_1} (24 - x_{21}) + \frac{p_1}{4p_2} = \frac{p_2}{p_1} x_{21} \quad \implies \quad x_{21} = 12 + \frac{p_1^2}{8p_2^2}$$

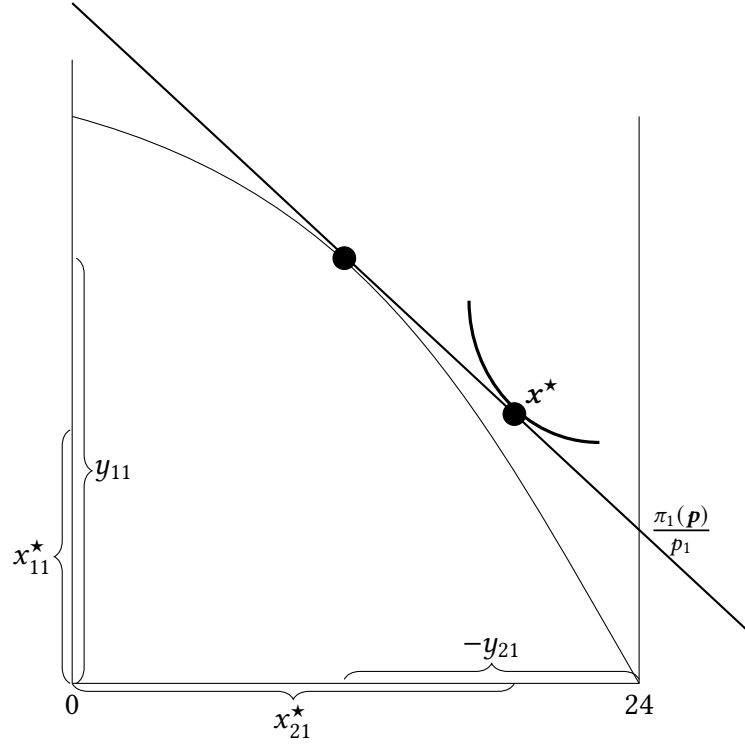


FIGURE 2.2: The consumer's optimal consumption bundle in the Robinson Crusoe economy.

From the budget constraint, demand of coconuts is then:

$$x_{11} = \frac{p_2}{p_1} \left(24 - \left(12 + \frac{p_1^2}{8p_2^2} \right) \right) + \frac{p_1}{4p_2} = 12 \frac{p_2}{p_1} - \frac{p_1}{8p_2} + \frac{p_1}{4p_2} = 12 \frac{p_2}{p_1} + \frac{p_1}{8p_2}$$

We show the consumer's problem graphically in Figure 2.2. Crusoe chooses the bundle that places him on the best indifference curve within his budget set. The budget line is given by $x_{11} = \frac{p_2}{p_1} (24 - x_{21}) + \frac{\pi_1(\mathbf{p})}{p_1}$. Its slope is $-\frac{p_2}{p_1}$ and the intercept above $x_{21} = 24$ is $\frac{\pi_1(\mathbf{p})}{p_1}$. Thus the budget line is the same as the firm's isoprofit function. It therefore passes through firm's choice (y_{11}, y_{21}) . An allocation and price vector $\mathbf{p}$ constitute a Walrasian equilibrium if markets clear in all goods when firms and consumers are optimizing subject to their constraints. In equilibrium, total demand for coconuts equals total supply:

$$\underbrace{12 \frac{p_2}{p_1} + \frac{p_1}{8p_2}}_{\text{Demand}} = \underbrace{\frac{p_1}{2p_2}}_{\text{Supply}} \implies \frac{p_1}{p_2} = \sqrt{32} = 2\sqrt{8}$$

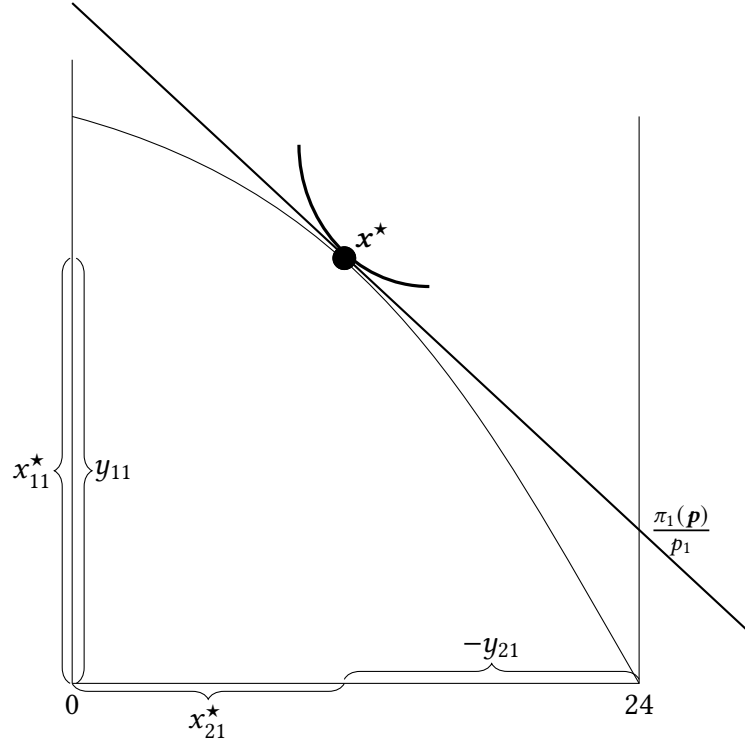


FIGURE 2.3: Walrasian equilibrium in the Robinson Crusoe economy.

At these prices, the market for leisure also clears:

$$x_{21} = \underbrace{24}_{\bar{\omega}_2} + y_{21} \implies \underbrace{12 + \frac{32}{8}}_{\text{Demand}} = \underbrace{24 - \frac{32}{4}}_{\text{Supply}} = 16$$

We show this graphically in Figure 2.3. Because the consumer's budget line and the firm's iso-profit line overlap, the ratio of prices that determine the slope of these lines $-\frac{p_2}{p_1}$ adjusts until markets clear where we have $x_{11} = y_{11}$ and $x_{21} = 24 + y_{21}$.

Thus we obtain the same result as the Pareto optimal outcome. This is not a coincidence and is as a result of the First Welfare Theorem, which is discussed in a later chapter.

In this economy, any price vector where $\frac{p_1}{p_2} = 2\sqrt{8}$ is an equilibrium. However, if we normalize one of the prices, for example, by setting $p_1 = 1$, then there is a unique equilibrium price vector in this economy.

Chapter 3

Partial Equilibrium

In this chapter we will consider another special case of the general economy defined in Chapter 2. The Robinson Crusoe example had $L = 2$ goods, but only one consumer and one firm ($I = J = 1$). Here we will consider a slightly more general case with $L = 2$ goods, but an arbitrary number of consumers and firms. We will consider a special case where preferences are such that we can focus our analysis on a single one of these goods, where we can think of the other good as representing “all other goods”. Under some assumptions on consumer preferences, we can find the market equilibrium price of this good in isolation. When we find the market equilibrium in a single good, we call it a *partial equilibrium analysis*.

A crucial assumption that we need to perform a partial equilibrium analysis, is that the good of interest has no wealth effects. A good having a wealth effect means that demand for it changes when a consumer’s wealth increases. For example, if someone’s wealth increases they might consume more high-quality food and less low-quality food. An example of a good that would arguably have no wealth effect is an eraser. If your income suddenly increased substantially, you would probably not change how many erasers you buy. But if erasers suddenly became very expensive, you might decide to buy fewer (and use a pen with correction fluid instead). More generally, if a good makes up a tiny share of your expenditure budget, wealth effects are less likely.

3.1 Model Setup

3.1.1 Consumers

Instead of using the $x_{\ell i}$ notation, we will call consumer i ’s consumption of the good of interest x_i , and their consumption of the other good m_i . We call this other good m_i the *numeraire*. Consumer

i 's utility function is:

$$u_i(m_i, x_i) = m_i + \phi_i(x_i)$$

where $\phi'_i(x_i) > 0$ and $\phi''_i(x_i) < 0$.